

CHAPTER 1

ADMINISTRATION

SECTION MC 101 GENERAL

101.1 Title. This code shall be known and may be cited as the “*New York City Mechanical Code*,” “NYCMC” or “MC.” All section numbers in this code shall be deemed to be preceded by the designation “MC.”

101.2 Scope. This code shall regulate the design, installation, maintenance, alteration and inspection of mechanical systems that are permanently installed and utilized to provide control of environmental conditions and related processes within buildings. This code shall also regulate those mechanical systems, system components, equipment and appliances specifically addressed herein. The installation of fuel gas distribution piping and equipment, fuel gas-fired appliances and fuel gas-fired appliance venting systems shall be regulated by the *New York City Fuel Gas Code*.

101.3 Intent. The purpose of this code is to provide minimum standards to safeguard life or limb, health, property, public welfare and the environment by regulating and controlling the design, construction, installation, quality of materials, location, operation and maintenance or use of mechanical systems.

101.4 Severability. If a section, subsection, sentence, clause or phrase of this code is, for any reason, held to be unconstitutional, such decision shall not affect the validity of the remaining portions of this code.

SECTION MC 102 APPLICABILITY

102.1 General. Where there is a conflict between a general requirement and a specific requirement, the specific requirement shall govern. Where, in a specific case, different sections of this code specify different materials, methods of construction or other requirements, the most restrictive shall govern.

102.2 Existing installations. Except as otherwise provided for in this chapter or elsewhere in this code, a provision in this code shall not require the removal, alteration or abandonment of, nor prevent the continued utilization and maintenance of, a mechanical system lawfully in existence on the effective date of this code.

102.3 Maintenance. Mechanical systems, both existing and new, and parts thereof shall be maintained in proper operating condition in accordance with the original design and in a safe and sanitary condition. Devices or safeguards that are required by this code shall be maintained in compliance with the applicable provisions under which they were installed.

102.3.1 Owner responsibility. The owner shall be responsible for maintenance of mechanical systems. To determine compliance with this provision, the commissioner shall have the authority to require existing mechanical systems to be inspected.

102.4 Additions, alterations or repairs. Additions, alterations, renovations or repairs to a mechanical system shall conform to requirements for a new mechanical system without requiring the existing mechanical system to comply with all of the requirements of this code. Additions, alterations or repairs shall not cause an existing mechanical system to become unsafe, hazardous or overloaded.

102.4.1 Minor additions, alterations, renovations and repairs. Minor additions, alterations, renovations and repairs to existing mechanical systems shall meet the provisions for new construction, unless such work is done in the same manner and arrangement as was in the existing system, is not hazardous and is approved.

102.4.2 Special provisions for prior code buildings. In addition to the requirements of Sections 102.4 and 102.4.1, the provisions of Sections 102.4.2.1 through 102.4.2.6 shall apply to prior code buildings.

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102.4.2.1 Fire and smoke dampers. In cases where the building's passive fire-resistance protection design, including rated construction, corridors and fire separations, complies with 1968 or prior codes, the determination as to whether a fire or smoke damper is required shall be permitted to be made pursuant to the *1968 Building Code*, or at the election of the applicant, the *New York City Mechanical Code*.

102.4.2.2 Guards and access to roofs and elevated structures. The provisions of Section 304.11 relating to guards and Section 306.5 relating to permanent means of access shall not apply where the equipment or appliances replace existing equipment or appliances in the same location.

102.4.2.3 Vibration isolators for cooling towers. Where a replacement cooling tower is installed and physical limitations prohibit compliance with the vibration isolator requirements of Section 313.3.7, such isolators may be omitted provided the devices shall comply with the *New York City Noise Control Code*.

102.4.2.4 Noncombustible fill for cooling towers. Where an existing exterior cooling tower with combustible fill within 15 feet (4572 mm) of the lot line is replaced, such replacement shall be permitted to have combustible fill, notwithstanding the provisions of Section 908.3.2.

102.4.2.5 Seismic supports. The determination as to whether seismic requirements apply to an alteration shall be made in accordance with the *1968 Building Code* and interpretations by the department relating to such determinations. Any applicable seismic loads and requirements shall be permitted to be determined in accordance with Chapter 16 of the *New York City Building Code* or the *1968 Building Code* and Reference Standard RS 9-6 of such code.

102.4.2.6 Wind resistance. Equipment, appliances and supports that are exposed to wind shall be designed and installed to resist the wind pressures determined in accordance with Chapter 16 of the *New York City Building Code*.

102.5 Change in occupancy. Refer to Chapter 1 of Title 28 of the *Administrative Code*.

102.6 Reserved.

102.7 Reserved.

102.8 Referenced standards. The standards referenced herein shall be those that are listed in Chapter 15 and in the rules of the department and such standards shall be considered as part of the requirements of this code to the prescribed extent of each such reference. Where differences occur between provisions of this code and the referenced standards, the provisions of this code shall apply. Refer to Article 103 of Chapter 1 of Title 28 of the *Administrative Code* for additional provisions relating to referenced standards.

102.8.1 Editions of referenced standards. References to standards in this code shall be to the editions of those standards provided for in Chapter 15 of this code, or as otherwise provided by rule.

102.9 Requirements not covered by this code. Requirements necessary for the strength, stability or proper operation of an existing or proposed mechanical system, or for the public safety, health and general welfare, not specifically covered by this code, shall be determined by the commissioner.

SECTION MC 103 DEPARTMENT OF BUILDINGS

103.1 General. Refer to the New York City Charter and Chapter 1 of Title 28 of the *Administrative Code*.

SECTION MC 104 DUTIES AND POWERS OF THE COMMISSIONER OF BUILDINGS

104.1 General. The commissioner shall have the authority to render interpretations of this code, adopt rules, and establish policies and procedures in order to clarify and implement its provisions. Such interpretations, policies, procedures, and rules shall be in compliance with the intent and purpose of this code. See the New York City Charter and Chapter 1 of Title 28 of the *Administrative Code* for additional provisions relating to the authority of the Commissioner of Buildings.

SECTION MC 105 APPROVALS

105.1 Approvals. Refer to Chapter 1 of Title 28 of the *Administrative Code*.

SECTION MC 106 PERMITS

106.1 General. Permits shall comply with this section, with Article 105 of Chapter 1 of Title 28 of the *Administrative Code*, and with requirements found elsewhere in this code.

106.2 Required. Any owner or authorized agent who intends to construct, add to, alter, repair, move, demolish, or change the occupancy of a building or structure, or to erect, install, add to, alter, repair, remove, convert or replace any gas, mechanical or plumbing system, the installation of which is regulated by this code, or to cause any such work to be done, shall first make application for construction document approval in accordance with Chapter 1 of Title 28 of the *Administrative Code* and this chapter and obtain the required permit.

106.3 Work exempt from permit. Exemptions from permit requirements of this code as authorized in Chapter 1 of Title 28 of the *Administrative Code* and the rules of the department shall not be deemed to grant authorization for any work to be done in any manner in violation of the provisions of this code or any other laws or rules.

106.4 Validity of permit. The issuance or granting of a permit shall not be construed to be a permit for, or an approval of, any violation of any of the provisions of this code or of any other law. Permits presuming to give authority to violate or cancel the provisions of this code or other law shall not be valid. The issuance of a permit based on construction documents and other data shall not prevent the commissioner from requiring the correction of errors in the construction documents and other data. The commissioner is also authorized to prevent occupancy or use of a structure where in violation of this code or of any other law.

106.5 Permits with respect to limited alteration applications. For permits with respect to limited alteration applications refer to Sections 28-101.5 and 28-104.6, Exception 1 of the *Administrative Code*.

SECTION MC 107 CONSTRUCTION DOCUMENTS

107.1 General. Construction documents shall comply with Article 104 of Chapter 1 of Title 28 of the *Administrative Code* and other applicable provisions of this code and its referenced standards. Such construction documents shall be coordinated with architectural, structural and means of egress plans.

107.2 Required documents. The applicant shall submit all of the documents specified in Sections 107.3 through 107.10 as appropriate to the nature and extent of the work proposed. Construction documents shall indicate the heating, ventilation, refrigeration, and other mechanical work to be performed, so drawn as to conform to the architectural and structural aspects of the building and to show in detail compliance with this code.

107.2.1 Composite plans. Composite plans showing compliance of architectural, structural, and mechanical parts of a building may be submitted provided that a clear understanding of each part is not impaired.

107.3 Lot diagram. The lot diagram shall be provided where applicable to the work proposed, including but not limited to the installation of exterior or rooftop equipment.

107.4 Building classification statement. Where applicable to the proposed work, the statement shall identify:

1. The occupancy group or groups that apply to parts of the building in accordance with Section 302 of the *New York City Building Code*;
2. The occupancy group of the main use or dominant occupancy of the building;
3. The construction type of the building in accordance with Section 602 of the *New York City Building Code*;
4. The structural occupancy/risk category in accordance with Table 1604.5 of the *New York City Building Code*;
5. The height of the building as defined in Section 202 of the *New York City Building Code*;

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6. The applicable measurements to the highest and lowest level of Fire Department access;
7. Whether the building is inside or outside of the fire districts; and
8. Whether the building is inside or outside a flood hazard area as such term is defined in Chapter 2 of the *New York City Building Code*.

107.5 Fuel-burning and fuel-oil storage equipment plans. Construction documents for fuel-burning and fuel-oil storage equipment shall contain plans that include the following data and information:

1. Diagrams of all distribution piping, including vent and fill piping for oil systems, and all safety cut-off and relief devices and valves in piping; indications of the sizes of distribution piping to be used and the fire-resistive ratings of the shafts or spaces containing distribution piping where required to be fire rated.
2. Diagrammatic floor plans showing the size, location, material for all fuel oil and transfer distribution piping and related equipment.
3. Floor plans or partial floor plans showing the location, operating pressure, layout, size, and listing information for all fuel-burning equipment, tanks, vents, and chimneys. The plans shall also indicate the method or means of providing air to the equipment space, including duct and opening sizes or means of direct venting.
4. Plans indicating the location and type of any relevant smoke and heat detectors, alarm and fire extinguishing systems.
5. Seismic protection and restraint details for piping and equipment as required by Chapter 16 of the *New York City Building Code*.
6. Details indicating the location, size and materials for all breechings; the thickness and type of insulation materials; and the clearances from combustible walls, partitions, and ceiling; and the fire-resistive ratings of rooms and spaces containing the equipment.
7. Details describing the type, material, listing information, height, and termination distances to adjacent properties and structures for chimneys and vents.
8. Details showing structural supports for fuel-burning equipment where required.
9. A statement as to the kind or grade of fuel to be used.
10. Plans indicating the location, arrangement, size, load, and maximum capacity of the burning, storage and fuel-pumping equipment.
11. In special flood hazard areas, construction documents shall comply with Appendix G of the *New York City Building Code*.

107.6 Heating systems. Construction documents for heating systems shall include the temperature to be maintained in every room and the output capacity in BTU per hour of the central heating source.

107.7 Boilers. Construction documents for boiler installations shall indicate the output capacity in BTU per hour, the operating weight of each boiler, the pressure setting of the relief valves, and such other data and information as required by this code.

107.8 Air-conditioning and ventilating systems. Construction documents for air-conditioning and ventilating systems shall contain plans that include the following data and information:

1. The location and sizes of all ducts; the location of all fire and smoke dampers, motors, fans, and filters; the type, air capacity, and size of all equipment; and where not shown on accompanying structural plans, the operating weight and manner of support of equipment.
2. The locations of smoke detecting devices.
3. The location and size of the fresh air intake, the design population, and the required ventilation for each room or space.
4. The amount of air to be exhausted or supplied from each outlet for each room or space.
5. In the case of ventilating or exhaust systems for ranges, fryers, ovens, and other similar types of restaurant or bakery equipment, for which a hood is required, the plans shall also show the type of extinguishing system, the location of heat detection devices, nozzles, piping, gas controls, manual and automatic control valves, method

of joining ducts, method and location of discharging exhaust from building, the location of break-glass controls, and the quantity in cfm designed for each hood.

6. The safety group classification of refrigerant utilized, if any.
7. The refrigerant concentration limit calculations and routing of all refrigerant piping for any air-conditioning system that contains more than 6.6 pounds (3.0 kg) of refrigerant. Refer to Chapter 11 for refrigerant piping requirements.

107.9 Refrigerating systems. Construction documents for refrigerating systems shall contain plans that include the following data and information:

1. The location of all machinery; the horsepower of compressors; the type and number of pounds of refrigerant to be used; and the air quantities for, and means of, ventilating the machinery space.
2. The location of refrigerant detectors and emergency switches for compressors and for ventilation in the machinery rooms.
3. The location of pressure relief piping and any city water connections and water-saving devices.
4. The tonnage capacity of the machine and the suction and discharge pressures at which the machine is rated.
5. The operating weight of the equipment.
6. The safety group classification of refrigerant utilized.
7. The refrigerant concentration limit calculations and routing of all refrigerant piping for any refrigerating system that contains more than 6.6 pounds (3.0 kg) of refrigerant. Refer to Chapter 11 for refrigerant piping requirements.

107.10 Energy efficiency. Construction documents shall include compliance documentation as required by the *New York City Energy Conservation Code*.

107.11 Retention of construction and submittal documents. Refer to Section 28-104.12 of the *Administrative Code*.

SECTION MC 108 INSPECTIONS AND TESTING

108.1 General. Except as otherwise specified, inspections required by this code or by the department during the progress of work may be performed on behalf of the owner by approved agencies or, if applicable, by special inspectors. However, in the interest of public safety, the commissioner may direct that any of such inspections be performed by the department. All inspections shall be performed at the sole cost and expense of the owner. Refer to Article 116 of Chapter 1 of Title 28 of the *Administrative Code* for additional provisions relating to inspections.

108.2 Required inspections and testing. In addition to any inspections otherwise required by this code or applicable rules, the following inspections shall be required:

1. Progress inspections.

- 1.1. Underground inspection and/or testing of installed piping, valves, fittings, support structures, anti-corrosion equipment and associated underground components shall be made before backfill is put in place. When excavated soil contains rocks, broken concrete, frozen chunks and other rubble that would damage or break the piping or cause corrosive action, clean backfill shall be on the job site ready for use in backfilling.

Exception: Ground-source heat pump loop systems tested and backfilled in accordance with Section 1210 shall be permitted to be backfilled prior to inspection.

- 1.2. Rough-in inspection shall be made after the roof, framing, fireblocking and bracing are in place and all ducting and other components to be concealed are complete, and prior to the installation of wall or ceiling membranes.
- 1.3. Inspections required by the *New York City Energy Conservation Code* shall be made in accordance with the rules of the department, as applicable.

2. Special inspections.

Special inspections shall be performed in accordance with this code and Chapter 17 of the *New York City Building Code*.

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3. **Final inspection.** Refer to Article 116 of Chapter 1 of Title 28 of the *Administrative Code*.

4. **Issuance of certificate of compliance.** Upon satisfactory inspection of service equipment and the satisfaction of all the requirements for sign-off, the department shall issue a certificate of compliance as applicable for the following service equipment:

- 4.1. Air-conditioning and ventilation systems,
- 4.2. Fuel-burning and fuel-oil storage equipment, including generators,
- 4.3. Refrigeration systems,
- 4.4. Heating systems, and
- 4.5. Boilers.

The requirements of Section 108.2 shall not be considered to prohibit the operation of any heating equipment or appliances installed to replace existing heating equipment or appliances serving an occupied portion of a structure provided that a request for inspection of such heating equipment or appliances has been filed with the department not more than 48 hours after such replacement work is completed, and all required inspections are performed before any portion of such equipment or appliances is concealed by any permanent portion of the structure.

108.2.1 Approved inspection agencies. Refer to Articles 114 and 115 of Chapter 1 of Title 28 of the *Administrative Code*.

108.2.2 Inspection of prefabricated construction assemblies. Prior to the approval of a prefabricated construction assembly having concealed mechanical work and the issuance of a permit, the department shall require the submittal of an evaluation report by an approved agency on each prefabricated construction assembly, indicating the complete details of the mechanical system, including a description of the system and its components, the basis upon which the system is being evaluated, test results and similar information, and other data as necessary for the commissioner to determine conformance to this code.

108.2.2.1 Test and inspection records. Required test and inspection records shall be available to the commissioner at all times during the fabrication of the mechanical system and the erection of the building; or such records as the commissioner designates shall be filed.

108.2.3 Exposure of work. The work shall remain accessible and exposed for inspection purposes. Neither the commissioner nor the city shall be liable for expense entailed in the removal or replacement of any material required to allow inspection.

108.3 Testing. Mechanical systems shall be tested as required in this code and in accordance with Sections 108.3.1 through 108.3.3. Tests shall be made by the permit holder and witnessed by the department or an approved agency.

108.3.1 New, altered, extended or repaired systems. New mechanical systems and parts of existing systems, that have been altered, extended, renovated or repaired, shall be tested as prescribed herein to disclose leaks and defects.

108.3.2 Apparatus, material and labor for tests. Apparatus, material and labor required for testing a mechanical system or part thereof shall be furnished by the permit holder.

108.3.3 Reinspection and testing. Where any work or installation does not pass an initial test or inspection, the necessary corrections shall be made to comply with this code. The work or installation shall then be resubmitted to the department for inspection and testing.

108.4 Sign-off of completed work. Refer to Article 116 of Chapter 1 of Title 28 of the *Administrative Code*.

108.5 Temporary connection. The commissioner shall have the authority to authorize the temporary connection of a mechanical system to the sources of energy for the purpose of testing mechanical systems or for use under a temporary certificate of occupancy.

SECTION MC 109 VIOLATIONS

109.1 General. Refer to Chapters 2 and 3 of Title 28 of the *Administrative Code*.

